

GHaLIB غالب : Generating Hope and Linguistic Irony in Banter

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Abstract

Hope speech is a new and growing topic in Natural Language Processing (NLP), but most of the research so far has focused on English. However, low-resource languages like Urdu are still ignored or underrepresented. We tried to solve this problem by using pre-trained multilingual models like XLM-RoBERTa, mBERT, EuroBERT and UrduBERT for detecting hope speech, especially in Urdu. We performed some initial basic text cleaning steps that include lower case transformation, normalization, and removal of stop words. We then proceeded to use the embeddings from these models and passed them through classifiers such as SVM, LightGBM and hyperparameter tuning were done to improve the results. We tested our approach on the PolyHope-M 2025 dataset benchmark, and the F1 score is taken as the performance metric. Our model gave a 95% F1 score for the Urdu binary classification and 63% for the Urdu multiclass classification. We also got excellent results in English, German, and Spanish. These results show that with proper fine-tuning and preprocessing, existing transformer models can work well even for low-resource languages like Urdu. The complete results and code implementations are available at <https://github.com/ahmedembeddedxx/GHaLIB/>.

1 Introduction

Sentiment analysis is an essential field in Natural Language Processing (NLP) that helps understand public sentiment across various domains, including social media, news, and customer feedback. However, analyzing sentiments involving sarcasm and hope speech presents a unique challenge due to the nuanced nature of human language. Sarcasm is particularly difficult to detect because it involves saying something that typically conveys the opposite of its literal meaning [1]. Many sentiment analysis models struggle with sarcasm, often misclassifying sarcastic statements as having a neutral or even positive sentiment when, in reality, they carry a negative undertone [2].

On the other hand, the speech of hope represents positive, motivational and uplifting communication. Detecting hope speech is crucial in social media analysis and mental health applications, as it can help identify and amplify positive discussions. However, hope speech often gets overshadowed by more dominant sentiments, making its detection a nontrivial task [3]. Our project aims to address these challenges by implementing advanced NLP techniques to accurately classify sarcasm and hope speech, contributing to a more comprehensive and fine-grained sentiment analysis system [4].

To understand the dataset characteristics, we first analyze the distribution of labels in both English and Urdu datasets. Figure 1 shows the class distributions for both languages, revealing that "Not Hope" is the most prevalent category in both cases, followed by generalized hope. The average text length and word count per sentiment category, shown in Figure 2, reveals interesting patterns across languages. In English, realistic hope expressions tend to be longer, while in Urdu, generalized hope statements are typically more verbose. Finally, word clouds in Figure 3 provide qualitative insights into the most frequent terms associated with each hope category. The English word clouds and Urdu examples reveal that realistic hope tends to include more concrete terms, while generalized hope contains more abstract concepts.

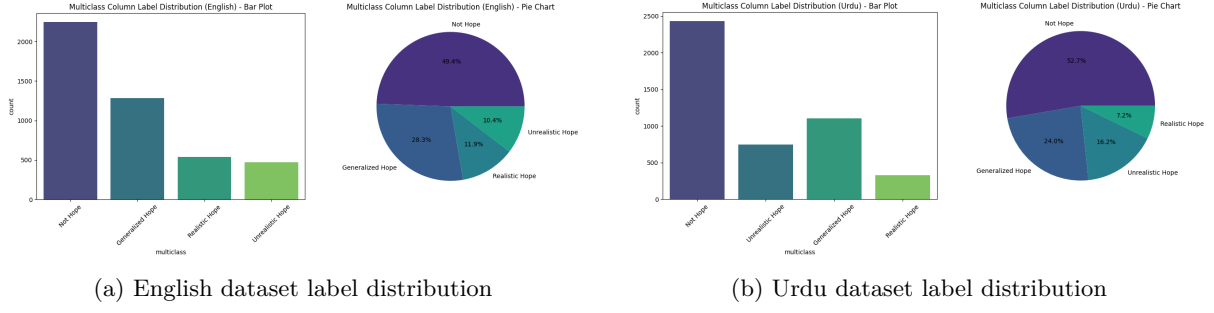


Figure 1: Class distribution in English and Urdu datasets showing the imbalance between categories, with "Not Hope" being the majority class in both cases (49.4% in English, 52.7% in Urdu).

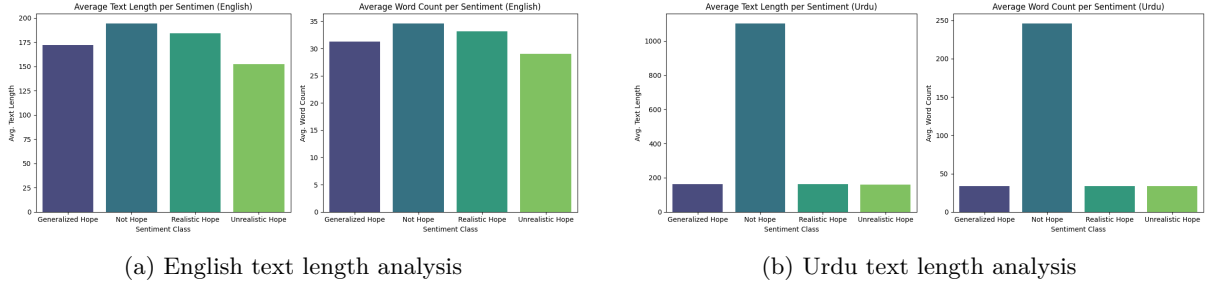


Figure 2: Average text length and word count per sentiment category for English and Urdu datasets. The variations suggest language-specific patterns in how different hope expressions are formulated.



Figure 3: Word clouds for Urdu hope speech categories showing characteristic vocabulary patterns for each sentiment class. The Arabic script terms highlight the unique linguistic features of Urdu hope expressions.



Figure 4: Word clouds for Urdu hope speech categories showing characteristic vocabulary patterns for each sentiment class.

2 Literature Review

2.1 Research on Pattern-Based Sarcasm Detection

Bouazizi et. al (2016) have been analyzing the detection of sarcasm on Twitter using a pattern-based approach [5]. They collected sarcastic tweets through a query to the Twitter API for tweets containing the hashtag *#sarcasm*, thus forming three balanced datasets with sarcasm being classified into three types: wit, whimper, and evasion. They used the classification with punctuation-related features, analyzing the contribution of punctuation patterns in sarcasm detection from given texts. Their proposed approach gave an F1 measure of about 82.7% on the Twitter dataset and an accuracy of about 83.46%, thus proving competitive with existing techniques for detecting sarcasm [5]. This study was further extended by Oxana et. al for context features, emotions and texts [6] [7].

2.2 Research on Low-Resource NLP

Dave and Desai et. al (2016) has handled a multilingual architecture for sarcasm detection with a focus on Hindi. They worked with the application of SVM classifier with 10-fold cross-validation with the simple Bag-of-Words approach for features and TF-IDF for frequency measurement [8]. The simplistic model classifies only 50% of the sarcastic sentences accurately, implying that the method is extremely insufficient and simple Bag-of-Words is not very effective in classifying sarcasm [7] [2]. They, therefore, added a model with an improved performance lexicon-based feature using both dictionary and corpus techniques to improve the model understanding of sarcasm in Hindi text [8]. Thus, this study is on proving the complexity of detecting sarcasm in a multilingual context while emphasizing the need for more advanced linguistic features than just using simple word frequency as representations [8] [9].

2.3 Research on Enhancing Sarcasm Detection with Contextual and Emotional Cues

Vitmana et al. (2023) proposed a sarcasm detection framework, integrating contextual, emotional, and sentiment-based features to enhance classification accuracy. The approach employs transformer-based models along with CNNs to capture linguistic incongruities that characterize sarcasm, thereby improving the training [6]. The framework achieves state-of-the-art performance over a variety of benchmarks, using advanced pre-trained sentiment and emotion analysis models. The findings reveal that emotion detection enhances sarcasm classification; also, combining contextual transformer embeddings with CNNs produces better results than legacy methods. This model performed far better on the Reddit, Internet Argument Corpus, and Twitter datasets, which further facilitates the place of sarcasm detection in focus [6] [9].

2.4 Research on Sentiment Analysis Using Deep CNNs

Zhao et al. (2017) tested analyzing tweets for sentiment using deep learning and their methodology tested on five different datasets to find out the performance. A deep Convolutional Neural Network (CNN) was used with a scoring mechanism called $SenScore(w)$ in which w refers to a lexicon word [9]. A tweet entered the model and was transformed into an $n \times k$ representation (with n is **number of words** and k is **embedding size**) and was then sent through three convolutional layers and a fully connected (FC) layer in that order [9]. GloVe-DCNN incorporates a regression model that integrates two major paradigms of value in sentiment analysis: one, a local context window, which focuses on words immediately near to it in an utterance, and the other, global matrix factorization, which recognizes a general relationship pattern of words in a huge body of text.

An average operational accuracy of 85.63% was obtained by this approach, beating others by at least 3.69% improvement, signifying that deep CNNs along with efficient word representations can act as boosters for the performance of sentiment analysis [9].

2.5 Research on ULMFiT and Transfer Learning for NLP Efficiency

Howard and Ruder et. al (2018) introduced the Universal Language Model Fine-tuning (ULMFiT), a method meant to extend transfer learning in NLP by using pre-trained language models. Conventional NLP models need a lot of labeled data and prove inefficient when trained from scratch, thus making transfer learning a vital element in boosting performance under limited-resource scenarios [4]. ULMFiT addresses these problems by utilizing several fine-tuning strategies including discriminative fine-tuning, where different layers of the model learn at different rates; slanted triangular learning rates, which allow fast convergence of the model; and gradual unfreezing, which has the effect of reducing overfitting by fine-tuning layers in a progressive manner. Their findings showed that pretraining on a general-domain corpus leads to dramatic performance improvements on downstream tasks and helps with overfitting and catastrophic forgetting-the two major challenges confronting transfer learning. ULMFiT was tested on six text classification datasets, where it outperformed the state-of-the-art methods by an error margin of 18-24% improvement [4].

The data-efficiency of the model was impressive, achieving competitive performances with as low as 100 labeled examples which matched the results of models trained on 100 times larger datasets. The study offers insight into the success of transfer learning approaches in the arena of NLP and underlines the role of fine-tuning strategies in improving a model’s generalization especially in low-resource settings [4] [9] [10].

2.6 Research on Multi-Task Learning for Sentiment Analysis

Tan et. al (2023) introduced a framework for multi-task learning in which a deeply layered BiLSTM neural network jointly performed sentiment analysis and sarcasm detection [11]. By simultaneously learning sentimental polarity and cues for sarcasm, the model minimizes mis-classification errors due to sarcasm. The results indicate that multi-task learning promotes efficiency, improving the score of the BiLSTM model by 3% as compared to techniques for classical sentiment analysis. Therefore, this approach has a state-of-the-art boost in sarcastic sentiment classification accuracy, resulting in even more stronger and interpretable sentiment analysis [11].

2.7 Research on Sentiment Analysis Using LLMs

Zhang and Deng et. al (2024) have studied how Large Language Models (LLMs) fare in sentiment analysis compared to Small Language Models (SLMs) that are trained on domain-specific datasets. They compared the models on 13 different tasks associated with 26 datasets for a final evaluation by sampling up to 500 examples from each test set for balanced evaluations. For the LLM part, they employed two Flan family models, Flan-T5 (XXL version, 13B) and Flan-UL2 (20B) [10], for the very reason that they have excellent zero and few-shot properties for their testing [12]. Their results state that LLMs, along with models like ChatGPT [13], perform excellently well in zero-shot settings and can handle sentiment-analysis tasks without prior fine-tuning on specific datasets [14].

Most importantly, LLMs, unlike SLMs, were very effective in few-shot learning contexts, which implies their incredible potential in cases of limited labeled training data. This shows that LLMs are highly

promising for real-world sentiment analysis applications, which involve expensive and few labeled data [12].

2.8 Research on Zero-Shot Classification

In Puri’s and Catanzaro’s (2019) investigation of generative language models for zero-shot text classification, natural language classifications were provided instead of task-oriented training examples. Such fine-tuning on title prediction tasks allowed GPT-2 [13] to generalize across benchmarks, and it was shown that zero-shot learning outperformed the random and majority class baselines. By reframing classification problems into multiple-choice ones, adaptability was improved; pretraining on an extensive scale using datasets endowed with metadata ensured good generalization [15].

2.9 Research on Prompt-based Classification

In 2023, Sun et al. proposed a framework, Clue And Reasoning Prompting (CARP), giving the clue identification, diagnosis reasoning, and decision-making process in order to enhance LLM-based text classification. Compared to existing methods of in-context learning, CARP produced superior results on elaborate linguistic tasks in low-resource settings while requiring far fewer labeled examples than traditional models [3].

2.10 Research on Implicit Sentiment Analysis

Xu et. al (2022) proposed KC-ISA, an implicit sentiment analysis model that integrates both knowledge enhancement and contextual features to assist in sentiment classification. Unlike sentiment analysis, which deals with the direct name of sentiment, implicit sentiment analysis requires a deep understanding of linguistic lingo and contextual understanding to make inferences [16][8]. To counter this challenge, the researchers designed a Knowledge Fusion Module which integrates external knowledge from XLORE. Using this external knowledge, the system is enabled to infer implicit sentiment expression in the absence of direct emotional clues [16] [7].

The KC-ISA model was evaluated on the SMP2019 implicit sentiment analysis dataset whereby these authors uncovered three key dependencies representing potential sources of improvement for implicit detection. The first, context dependency, means that a text surrounding a given sentiment would determine whether the sentiment is positive or negative; whilst sentiment target dependency keeps track of whether a sentiment has been assigned to the correct target; finally, knowledge and common sense dependency exploit knowledge concerning such external objects which could help in qualifying the sentiment [14]. The integration of dependency factors is what made the KC-ISA model highly efficiently, and it achieved an accuracy of 0.786 and F1 of 0.755 for implicit sentiment classification [16].

2.11 Research on RGPT for LLM-Based Classification

Zhang et. al (2024) proposed RGPT, which works similar to an adaptive boosting framework for text classification-the LLMs are fine-tuned iteratively while at the same time the distributions of the training samples are modified. The RGPT builds a highly specialized classifier by the ensemble of several fine-tuned models [17]. It goes beyond the performance of eight strong contenders PLM and seven versions of LLMs across four benchmarks and above-average human classification performance. The adaptive boosting technique generalizes better by focusing misclassified samples and therefore significantly enhances LLM-based text classification [17].

3 Methodology

This study developed a robust text classification model to address a binary classification task, prioritizing the reduction of false negatives (FN, 6%) and false positives (FP, 10%). We selected xlm-roberta-base as the primary model after comparative experiments with xlm-roberta-large and eurobert, a multilingual BERT variant. The base model outperformed its counterparts in accuracy and F1-score while requiring fewer computational resources, making it suitable for deployment on a 16GB NVIDIA T4 GPU in the Kaggle environment.

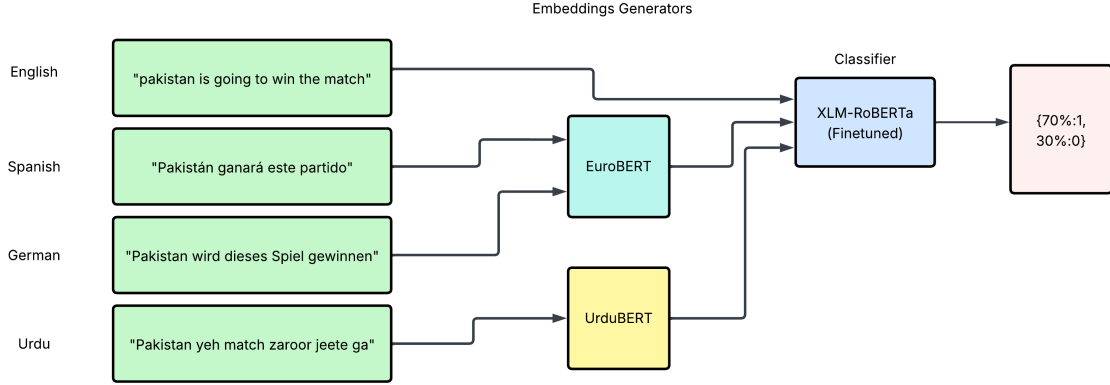


Figure 5: Architecture of GHaLIB

The dataset, comprising text samples labeled as positive or negative, was preprocessed using the xlm-roberta-base tokenizer with a maximum sequence length of 128 tokens. Data was split into training (70%), validation (15%), and test (15%) sets to ensure robust evaluation. To mitigate class imbalance and reduce FN, we implemented a custom WeightedTrainer with a cross-entropy loss function, assigning a 1.5x higher weight to the positive class. This adjustment encouraged the model to prioritize correct positive predictions, critical for applications where missing positives is costly.

Hyperparameter optimization was performed using Optuna, an efficient framework for automated tuning. We conducted 30 trials, exploring a comprehensive search space: learning rates (5e-6 to 5e-5), batch sizes (4, 8, 16), warmup ratios (0.0 to 0.3), weight decays (0.0 to 0.1), and dropout rates (0.1 to 0.3). The optimization objective maximized a weighted F1-score while penalizing FN (0.5 weight) and FP (0.3 weight), balancing overall performance with error reduction. Gradient accumulation (8, 4, or 2 steps for batch sizes 4, 8, 16, respectively) maintained an effective batch size of 32, and gradient checkpointing alleviated memory constraints. Training incorporated early stopping (patience=2), cosine learning rate scheduling, and label smoothing (0.1) to enhance generalization and reduce overconfidence.

Post-training, we tuned the classification threshold (0.3 to 0.8) on the validation set to optimize the FN-FP trade-off. The model was evaluated using accuracy, weighted F1-score, FN rate, and FP rate, with results visualized through confusion matrices and performance curves (loss, accuracy, F1, FN/FP rates) at 300 DPI using Times New Roman font. All experiments were conducted in Python with the Hugging Face Transformers library, ensuring reproducibility and scalability.

4 Results & Experimentation

In this study, we were able to perform a detailed evaluation of the multilingual transformer-based architectures in the context of binary and multiclass text classification tasks across four languages: English, German, Spanish, and Urdu. Our main focus was to determine the best-performing architecture when fully fine-tuned. Out of those tested on various tasks, XLM-RoBERTa-Base was found to be the top model performing for all the scenarios, achieving the highest accuracy scores on both binary and multiclass tasks.

XLM-RoBERTa-Base is a transformer model based on the framework of RoBERTa, trained on a very large and diverse multilingual corpus covering over 100 languages. XLM-RoBERTa-Base is primarily designed to allow the deep capture of contextual information across languages, which is very handy for cross-lingual, multipurpose understanding tasks. XLM-RoBERTa-Base provides a fair balance in performance and computational efficiency during the complete fine-tuning process, in contrast to the larger version—XLM-RoBERTa-Large [2].

For instance, it was recorded that XLM-RoBERTa-Base achieved a high 94.5% accuracy rate for Urdu, along with 87.2%, 86.3%, and 84.5%, respectively, for German, English, and Spanish, marking the highest result recorded across all language/settings when it came to binary classification tasks. The results prove that it could generalize remarkably across different languages, while handling varying scripts and linguistic layouts.

Table 1: Performance metrics

Sr. No.	Preprocessing	Architecture	Model	Type	Language	Acc.	F-1 (D)
1	Basic + SS	RoBERTa	ADB + SVM + LGB	Bin	English	0.81	0.81
2	Basic + SS	RoBERTa	ADB + SVM + LGB	MC	English	0.68	0.56
3	Basic + SS	RoBERTa Urdu	ADB + SVM + LGB	Bin	Urdu	0.93	0.93
4	Basic + SS	RoBERTa Urdu	ADB + SVM + LGB	MC	Urdu	0.77	0.66
5	Basic + SS	RoBERTa + ROS	ADB + SVM + LGB	MC	English	0.61	0.52
6	Basic + SS	RoBERTa + ROS	ADB + SVM + LGB	MC	Urdu	0.56	0.24
7	Basic + SS	DistilBERT	ADB + SVM + LGB	Bin	English	0.67	0.54
8	Basic + SS	DistilBERT	ADB + SVM + LGB	MC	English	0.78	0.78
9	Basic + SS	GloVe	ADB + SVM + LGB	Bin	English	0.73	0.73
10	Basic + SS	GloVe	ADB + SVM + LGB	MC	English	0.62	0.44
11	Basic + SS	UrduBERT	ADB + SVM + LGB	Bin	Urdu	0.93	0.93
12	Basic + SS	UrduBERT	ADB + SVM + LGB	MC	Urdu	0.71	0.51
13	Basic + SS	RoBERTa	ADB + SVM + LGB	Bin	English	0.85	0.85
14	Basic + SS	RoBERTa	ADB + SVM + LGB	MC	English	0.68	0.56
15	Basic + SS	RoBERTa Urdu	ADB + SVM + LGB	Bin	Urdu	0.93	0.93
16	Basic + SS	RoBERTa Urdu	ADB + SVM + LGB	MC	Urdu	0.73	0.53
17	Basic + SS	MultiLing BERT	ADB + SVM + LGB	Bin	English	0.76	0.76
18	Basic + SS	MultiLing BERT	ADB + SVM + LGB	MC	English	0.68	0.49
19	Basic + SS	MultiLing BERT	ADB + SVM + LGB	Bin	Urdu	0.93	0.93
20	Basic + SS	MultiLing BERT	ADB + SVM + LGB	MC	Urdu	0.73	0.50
21	Basic + SS	Finetuned RoBERTa	-	MC	English	0.75	0.70
22	Basic + SS	Finetuned RoBERTa	-	Bin	English	0.85	0.85
23	Basic + SS	XLM RoBERTa	-	Bin	English	-	0.86
24	Basic + SS	XLM RoBERTa + EuroBERT	-	Bin	German	-	0.87
25	Basic + SS	XLM RoBERTa + EuroBERT	-	Bin	Spanish	-	0.84
26	Basic + SS	XLM RoBERTa + UrduBERT	-	Bin	Urdu	-	0.94
27	Basic + SS	XLM RoBERTa	-	Multi	English	-	0.71
28	Basic + SS	XLM RoBERTa + EuroBERT	-	Multi	German	-	0.70
29	Basic + SS	XLM RoBERTa + EuroBERT	-	Multi	Spanish	-	0.68
30	Basic + SS	XLM RoBERTa + UrduBERT	-	Multi	Urdu	-	0.65

For multi-class classification, it still managed to pull in very impressive figures compared to other languages, with 71.2% for English, 70.13% for German, 68.6% for Spanish, and the lowest, which is still high in comparison to the rest, at 65.3% for Urdu. Model versatility and robustness emerged, as its performance remained superior despite the challenging complexity of multi-class classification tasks.

5 Conclusion

In the binary classification experiments, XLM-RoBERTa-Base obtained a 94.5% accuracy in Urdu, the highest among all languages and settings, followed by 87.2% in German, 86.3% in English, and 84.5% in Spanish. The results thus show its ability to generalize with languages differing in scripts and structures.

In multi-class classification, the model once again retained a superior position, with the performance results of 71.2% in English, 70.13% in German, 68.6% in Spanish, and 65.3% in Urdu. The results of the model continued to exhibit a high performance despite the added complication in multi-class, thus confirming its ability and robustness.

Overall these findings emphasize the might of XLM-RoBERTa-Base as a trustworthy and strong contender for multilingual text classification. Its consistent cross-domain effectiveness strengthens justifiability as an option for any real-world multilingual NLP applications, primarily when considering computational resources.

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